MILITARY SPECIFICATION

PALLET, SPECIAL METAL FOR STINGER WEAPON SYSTEM

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope: This specification covers the requirements and quality assurance provisions for the manufacture of special metal pallets for use in unitizing ammunition (see 6.1).

1.2 Classification. The special metal pallets shall be of the following type (see 6.1):

Type I - Pallet, Special Size 66.44 inches x 45 25 inches, Sheet Metal

2 APPLICABLE DOCUMENTS

2 1 Government documents

2.1 1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

TT-C-490		Cleaning and Preparation of Ferrous and Coated Surfaces for Organic Protective Coatings

MILITARY

MIL-W-12332	-	Welding Resistance, Spot, Seam and Production, for Fabricating Assemblies of Low Carbon Steel
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-70625	-	Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval of

STANDARDS

FEDERAL

FED-STD-101	-	Test Procedures for Packaging Materials

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-1235	-	Single and Multi Level Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1261	-	Welding Procedures for Constructional Steel
MIL-STD-1660	-	Design Criteria for Ammunition Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

ACV00193	C.O.A.I (Combination of Adopted Items) Pallet and Pallet Adapters Unitization of PA158 container
ACV00194	Top Assembly, Pallet Adapters PA158 container
ACV00195	Bottom Assembly, Pallet Adapter PA158 container
ACV00196	Pallet Special size 66.44"x 45.25", Sheet Metal

(Copies of other Government documents, drawings and publications required by contractors in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government documents . The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed ir the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103-1167.)

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AMERICAN WELDING SOCIETY

ANSI/AWS B4.0 - Mechanical Testing of Welds Standards Methods for

(Application for copies of AWS publications should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, Florida 33135).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Pallets. The pallet assemblies shall comply with all requirements specified on Drawing ACV00193 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Stacking strength. The pallets shall withstand a 20,000 pound minimum (min.) stacking load (see 4.5.1)

3.4 Welding. The welding shall comply with the requirements of MIL-W-12332 or MIL-STD-1261 and weld callout in the applicable drawings.

3.5 Weld integrity.

3.5.1 Visual inspections. The load bearing surface welds (see applicable dwg.) of the pallet assembly shall be free of the following defects

a. Undercut.

b. Overlap.

c. Surface cavities.

d. Surface cracks in weld metal or in heat affected zone of base metal.

e. Lack of weld penetration.

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In addition, fine scattered (average pore diameter equal to or less than 0.030 inches) porosity exceeding 13 pores in any one square inch shall be rejected. Not more than two pores in any one square inch of weld face shall be rejected. No single pore may exceed 0.060 inches in diameter. Linear surface porosity in excess of 0.500 inches in or in excess of 25 percent of total length, whichever is less shall be rejected.

3.5.2 Weld coupon bend test. The guided bend specimens shall have no open discontinuities exceeding 3.2mm (1/8 in.) measured in any direction on the convex surface of the specimen after bending, except that cracks occurring on the corners shall not be considered, unless there is definite evidence that they result from slag inclusions or other internal discontinuities (see 4.5.2)

3.6 Painted surface. The painted surface shall comply with 3.6.1 or 3.6.2, as applicable.

3.6.1 Conventional coating. The painted surfaces of the pallets shall comply with the requirements of the applicable drawings, TT-C-490 and with instructions in MIL-STD-171.

3.6.2 Chemical agent resistant coating (CARC). The painted surfaces of the pallets shall comply with the requirements of Dwg. ACV00193 and applicable drawings.

3.7 Unit load test. The unit loaded pallets shall conform to the requirements specified in MIL-STD-1660 (see 4.5.3).

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship.

3.9.1 Pallets. The pallets shall be regular, smooth, free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability and safety of the pallets. Weld splatter is permissible on non-load bearing surfaces provided it is firmly attached.

3.9.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided there are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total

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coverage. Paint chips which expose primer or bare metal are acceptable provided that are not more than 0.25 inches in diameter and no more than two such paint chips on a pallet. Scratches which expose primer or bare metal are permissible provided they are no more than 1.5 inch in length and no more than 10 inches total on a pallet. Bare metal on contact surfaces on bottom of skids and in areas inaccessible to paint is permissible.

3.9.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4)

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Drawings	Part Description	Quantity
AC200000429	Post	6
ACV00193	Pallet Assemblies, Unpainted	10
ACV00193	Pallet Assemblies, Painted	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Assemblies and Components	1 OF 1	As applicable
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Post (Dwg. AC200000429)			
	Height		3.2	Gage
	Evidence of poor workmanship		3.9	Visual
	Material thickness		3.2	Gage
	Pallet Assemblies, Unpainted			
	Examination for defects		3.2	4.4.2.1
	Pallet Assemblies, Painted			
	Examination for defects		3.2	4.4.2.2
	Stacking strength		3.3	4.5.1
	Unit load tests		3.7	4.5.3

NOTES

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc , may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4 2.1	Unpainted Pallet Assembly		ACV00193
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Mating surface of weld incorrect 1/	LEVEL IV	3.5.1	Visual
102	Location of adapter mounting hole(s) in deck	LEVEL IV	3.2	Gage
103	Location of adapter mounting hole(s) in skid	LEVEL IV	3.2	Gage
104	Height of pallet	LEVEL IV	3.2	Gage
105	Overall length	LEVEL IV	3.2	Gage
106	Overall width	LEVEL IV	3.2	Gage
107	Weld without wrap 1/	LEVEL IV	3.2	Visual
108	Visual inspection of weld, unsatisfactory 1/	LEVEL IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	LEVEL V	3.9.1, 3.9.3	Visual

NOTES: 1/ Pertain to load bearing welds. These are the welds which join the post to the skid and the post to the deck

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Pallet Assembly (Painted)		ACV00193
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Protective coating thickness (total thickness minus primer thickness (see Dwg. AC200000423 and 1/ and 2/))	LEVEL IV	3.2	Gage
102	Stacking strength	LEVEL IV	3.3	4.5.1
Minor				
201	Protective finish missing, incomplete or incorrectly applied	LEVEL V	3.9.2	Visual
202	Marking incorrect or missing	LEVEL V	3.2	Visual
203	Evidence of poor workmanship (other than protective finish)	LEVEL V	3.9	Visual

NOTES 1/ This inspection only applies when the assembly has been painted with CARC.
2/ The primer thickness shall be determined from a coupon test panel.

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provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.4.3 Testing.

4.4.3.1 Weld coupon bend test. (See 3.5.2), Major defect - During the first half of the production shift and during the last half of the production shift, two weld coupons shall be prepared by the welder in accordance with ANSI/AWS B4.0. If any test coupon fails to meet the requirement, when tested in accordance with 4.5.2, the quantity of pallets welded during that shift and after the earlier weld coupon test (if applicable) shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in Table I, 4.4.2.1, 4.4.2.2 and in 4.5. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. When Automated Acceptance Inspection Equipment is to be used, the provision of MIL-A-70625 shall apply. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Stacking strength. The pallet shall be stacked with the required stacking load for a period of one (1) hour minimum. After that time, the pallet shall not have any evidence of breaking, bending, buckling or failure of any of the welds. Any pallet which fails to meet the requirement shall be classed defective.

4.5.2 Weld coupon bend test. A guided bend test, as specified in ANSI/AWS B4.0, shall be performed on the coupons. Any coupon which fails to meet the requirement shall be classed defective.

4.5.3 Unit load test. (First article inspection only) Prototype unit loads for the pallets shall be fabricated in accordance with MIL-STD-1660 with the following exceptions.

The gross weight of the ammunition load shall be 1,090 pounds-50 pounds, evenly distributed over the deck and shall obtain a height of 38 inches ± 2 inches. The following tests, on the loaded pallets, shall be performed in accordance with MIL-STD-1660.

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4.5.3.1 Repetitive shock test. The acceptance/rejection criteria specified in section 7 of Method 5019.1 in FED-STD-101 is mandatory.

4.5.3.2 Edgewise drop (rotational) test. The acceptance/rejection criteria specified in section 7 of Method 5008.1 in FED-STD-101 is mandatory.

4.5.3.3 Incline impact test.. The acceptance/rejection criteria specified in section 7 of Method 5023 in FED-STD-101 is mandatory.

4.5.3.4 Sling compatibility test. The acceptance/rejection criteria specified in section 5 of MIL-STD-1660 shall apply.

5. PACKAGING

5.1 Preservation and packing. None required.

5.2 Packing.

5.2.1 Commercial. The metal pallet shall be packed in accordance with ASTM-D3951.

5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM D3951-82 and the following additional instructions. No shipping marking shall be placed directly on the pallets. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If pallets are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

5.4 Shipping. When pallets from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

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6. NOTES

6.1 Intended use. This metal pallet is intended for use with Stinger Basic/Post/RMP Guided Missile System FIM92A/92B/92C/92D and other applicable pallet adapters.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of contractor inspection equipment designs for approval. See MIL-A-48078. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAR-R, Picatinny Arsenal, NJ 07806-5000 for evaluation/approval by the technical activity responsible for the preparation of this specification.

6.5 Subject term (key word) listing.
Ammunition
Metal
Pallet

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A974)

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-P-71171 (AR)	930405

3 DOCUMENT TITLE

PALLET, SPECIAL METAL FOR STINGER WEAPON SYSTEM

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c ADDRESS (Include Zip Code) ATTN. SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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